

7.1 Scaling Laws

Loss falls as a power of compute, data, and parameters — until you starve one of them.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Bigger models, trained on more data with more compute, tend to get better at next-token prediction.

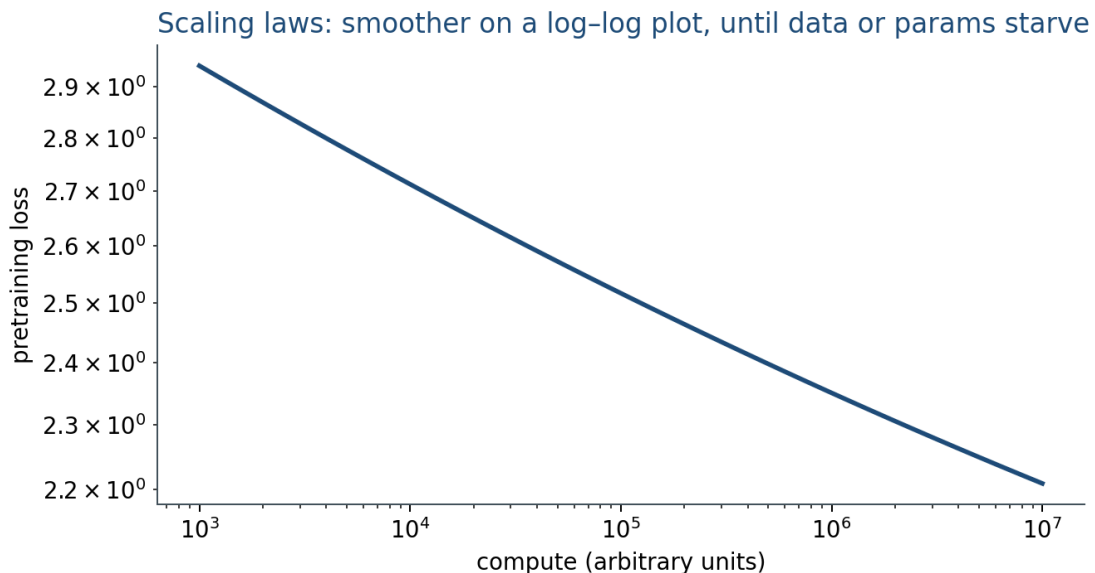
Scaling laws make that trend a curve you can fit, not a slogan. They do not promise a better chatbot; they promise a lower **pretraining loss** if the recipe stays the same.

1. Kaplan: loss as a power of compute

Kaplan et al. (2020) fit language-model cross-entropy against compute C (and against parameters N , dataset size D) and saw smooth **power laws** over many orders of magnitude:

$$L(C) \approx a C^{-b} + L_{\infty}.$$

The irreducible term L_{∞} is entropy of the data plus whatever the architecture cannot represent. The interesting bit is the slope $-b$: extra FLOPs still buy loss, with diminishing returns.

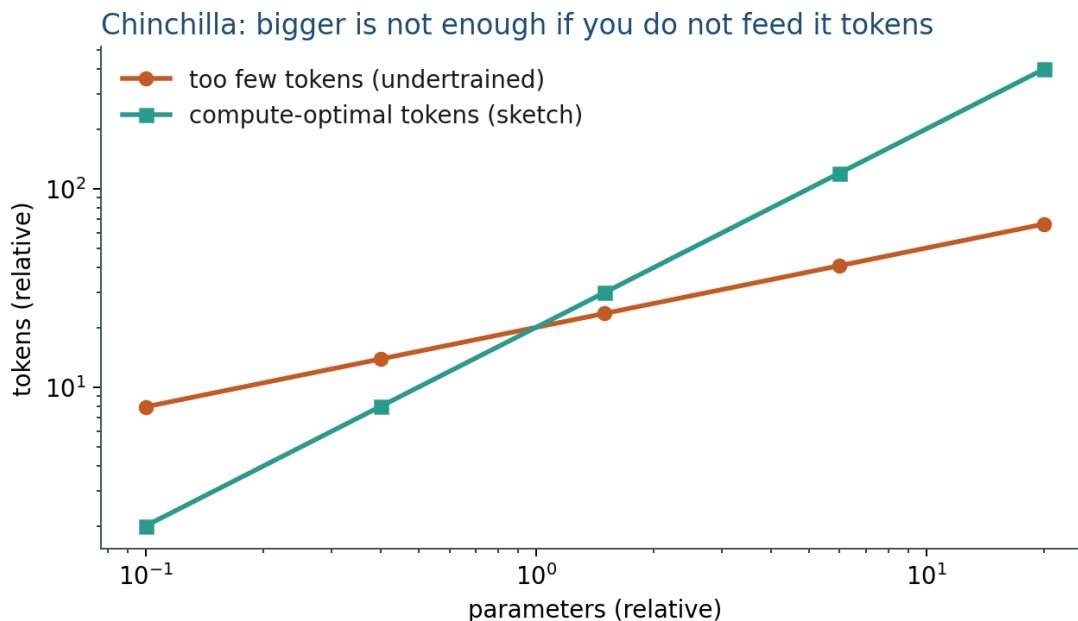


Fix two of $\{N, D, C\}$ and the third is constrained. Early GPT-3-style runs were **compute-optimal for a short training budget** that underused data: a large N , not enough tokens.

2. Chinchilla: tokens versus parameters

Hoffmann et al. (2022), **Chinchilla**: for a fixed FLOP budget, you should scale **tokens and parameters together**, not dump the budget into a huge model that sees the data once. Their fit said, roughly,

equal scaling and about **20 tokens per parameter** for the models they trained.



A smaller model trained longer can beat a larger undertrained one. That is why later LLaMA-style recipes look “small” next to GPT-3 and still read well: they were fed more tokens.

3. What a law does not tell you

Downstream accuracy is not L . Alignment (Weeks 8–9), data quality, mixture, and context length all sit off the curve. Inference cost scales with N even if you already paid for training. For a project: cite the checkpoint size and the token budget if you know them; do not claim “we scaled” because you trained two extra epochs on a toy set. Lab 7 fits $L = aC^{-b} + c$ on constructed points so you see the algebra.

4. Practice

1. Two models, same FLOPs. One has $4 \times$ the parameters and $1 / 4$ the tokens of the other. Who does Chinchilla pick, and why?
2. If L_∞ is already almost your measured loss, what does extra compute buy?
3. Why can a scaling law look great while a chat eval is flat?